SOFTWARE ENGINEERING&DESIGN PRINCIPAL LAB

EXPERIMENT 2

Identification of Problem, Objectives, Scope and Feasibility Analysis

Experiment No.	2
Project Title	Smart Traffic Management System
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1. Problem Identification

1.1 Background / Context

The project is developed in the Transportation and Intelligent Transportation Systems (ITS) domain. Urban roads commonly experience congestion at intersections because traffic demand changes throughout the day, while conventional traffic signals may operate using fixed timing plans. A software-based smart traffic management system can collect traffic information, analyze traffic density and provide data-driven signal-timing decisions. The proposed project focuses on a manageable software implementation for multiple intersections using simulated or recorded traffic data rather than requiring physical traffic-control infrastructure.

1.2 Existing Problem

Traffic operators and road users face delays, long queues and inefficient use of road capacity when signal timings do not adequately reflect current traffic conditions. Traffic density can vary considerably between approaches of the same intersection and across different time periods. Without a centralized monitoring and decision-support system, operators may have limited real-time visibility of congestion and may need to rely on predefined signal schedules or manual observation.

1.3 Existing Method / Current System

A conventional approach uses predetermined traffic-signal cycles and manual monitoring by traffic personnel. Traffic conditions may be observed through cameras, field personnel or other available sources, while signal plans are adjusted manually or according to predefined schedules. In a small software prototype, traffic conditions are typically represented through manually entered or static datasets and are not automatically converted into optimized signal decisions.

1.4 Limitations of the Existing System

- Fixed signal timings may not respond effectively to changing traffic density.
- Traffic information may be distributed across different observations or systems rather than presented in one dashboard.
- Manual monitoring and intervention can increase response time during congestion.
- Historical traffic information may not be readily available for comparison and analysis.
- There is limited automated support for prioritizing heavily congested approaches or special traffic situations.

2. Problem Statement

Traffic authorities and traffic-control operators need to manage congestion at road intersections where traffic demand varies continuously. Existing fixed-timing and manually monitored approaches may not respond quickly to changing traffic density, resulting in unnecessary waiting time, longer queues and inefficient signal utilization. A software-based Smart Traffic Management System is therefore proposed to monitor intersection-wise traffic conditions, analyze traffic density, generate adaptive signal-timing recommendations, highlight congestion and provide historical analytics through a centralized dashboard. The initial system will use simulated or recorded traffic data so that the solution remains feasible within the resources and scope of a student software project.

3. Objectives of the Project

S. No.	Objective
1	To develop a centralized dashboard for monitoring traffic conditions at multiple intersections.
2	To collect, store and display traffic-density

	information for different road approaches.
3	To analyze traffic density and determine suitable signal timings using a rule-based optimization algorithm.
4	To identify and highlight congested intersections or approaches for traffic operators.
5	To provide historical traffic statistics and reports for identifying recurring congestion patterns.
6	To provide a mechanism for prioritizing emergency-vehicle or high-priority traffic situations within the software model.
7	To reduce unnecessary waiting time and improve utilization of signal green time through adaptive recommendations.
8	To provide a user-friendly and maintainable software system that can be extended with advanced prediction features.

4. Scope of the Project

4.1 Target Users / Stakeholders

S. No.	User / Stakeholder	Role
1	Traffic Control Operator	Monitor traffic conditions and review signal recommendations.
2	Traffic Management Administrator	Manage intersections, signal configurations, users and system settings.
3	Traffic Planning / Analysis Staff	Review historical traffic statistics and congestion reports.
4	Development / Maintenance Team	Maintain the application, database and traffic-analysis logic.

4.2 Functional Scope

- User authentication and role-based access for authorized operators and administrators.
- Management of intersection, road-approach and traffic-signal information.

- Entry/import of simulated or recorded traffic-density data.
- Dashboard showing current traffic density, signal status and congestion level for selected intersections.
- Rule-based adaptive signal-timing recommendation based on traffic density and configurable constraints.
- Congestion and high-density alerts for traffic operators.
- Historical traffic analysis, charts and report generation.
- Emergency-vehicle priority workflow within the software simulation.

4.3 Data Scope

S. No.	Data / Information	Description
1	User and Role Data	User accounts, roles and access permissions.
2	Intersection Data	Intersection ID, name/location label, approaches and configuration.
3	Traffic Density Data	Vehicle counts/density values by approach, timestamp and traffic level.
4	Signal Data	Signal phase, current status, green/red duration and configurable timing limits.
5	Congestion and Alert Data	Congestion level, alert type, timestamp and acknowledgement status.
6	Historical Analytics Data	Stored traffic observations used for trends, comparisons and reports.

4.4 Out of Scope

- Physical installation or direct control of real-world traffic lights and roadside controllers.
- Live deployment using city-wide CCTV, IoT sensors or proprietary traffic-control infrastructure.
- Automatic enforcement of traffic violations, such as issuing fines or identifying offenders.
- Fully autonomous city-wide traffic prediction and optimization using production-grade AI models.

5. Technical Feasibility Analysis

The proposed system is technically feasible as a web-based software prototype because its core requirements—web interface, server-side APIs, relational data storage, traffic-density processing and dashboard visualization—can be implemented using commonly available development technologies. The initial implementation does not depend on specialized physical hardware.

5.1 Proposed Technology Stack

Component	Proposed Technology
Frontend	React.js with HTML5/CSS3 and responsive UI
Backend	Node.js with Express.js REST API
Database	MySQL
Programming Language(s)	JavaScript / SQL
Development Tools/IDE	Visual Studio Code, Git and GitHub
Other Technologies	REST APIs, Charting Library, JSON, browser-based dashboard

5.2 Hardware Requirements

Hardware	Requirement
Processor	Dual-core processor or better; modern student laptop/desktop is sufficient.
RAM	Minimum 8 GB recommended.
Storage	At least 10 GB free space for development tools, source code and database.
Other Hardware	Internet/network access for package installation and collaborative development; no specialized traffic hardware required.

5.3 Software Requirements

Software	Version / Requirement
Operating System	Windows 10/11, Linux or macOS
IDE / Editor	Visual Studio Code or equivalent
Database	MySQL 8.x or compatible current version
Framework / Libraries	React.js, Express.js, Node.js, charting library and standard npm packages
Other Software	Git, GitHub account, modern web browser and Node.js package manager

5.4 Technical Skills Required

- Frontend development using React.js, HTML, CSS and JavaScript.
- Backend/API development using Node.js and Express.js.

- Relational database design and SQL using MySQL.
- Traffic-density analysis and rule-based signal-timing algorithm design.
- Software testing, debugging, Git-based version control and basic deployment practices.

Technical Feasibility Assessment: The required development software, general-purpose hardware and programming skills are available or learnable by a four-member student development team. The project can be implemented and demonstrated using simulated/recorded traffic data, avoiding dependence on specialized physical traffic infrastructure.

Conclusion:

☑ **Technically Feasible**

☐ **Technically Not Feasible**

Justification: The proposed technology stack is mature and accessible, the hardware requirements are modest, and the main traffic-management functions can be implemented as software modules. Specialized real-world traffic controllers and sensors are intentionally excluded from the current scope.

6. Operational Feasibility Analysis

6.1 User Acceptance

The intended users—traffic-control operators, administrators and analysis staff—can benefit from a centralized interface that presents traffic conditions and recommendations in a structured manner. Acceptance is expected to be favorable because the system supports existing monitoring and decision-making rather than requiring users to replace their role with a fully autonomous system.

6.2 Ease of Use

The dashboard will use clear navigation, intersection cards/maps, traffic-density indicators, signal-status displays, alerts and charts. Common actions such as selecting an intersection, viewing congestion, checking signal recommendations and generating reports will be kept simple.

6.3 Existing Workflow

The proposed system fits into the existing workflow by providing a centralized digital layer for monitoring and analysis. Instead of relying only on manual observation or fixed

schedules, an operator can view traffic conditions, review the system-generated recommendation and take an informed decision. The prototype can demonstrate this workflow without directly controlling real-world signals.

6.4 Training Requirements

Only basic training is expected. A short user guide or demonstration can explain login, dashboard navigation, intersection monitoring, alert handling, signal recommendation review and report generation. Administrators may require additional training for configuration and user management.

6.5 Expected Operational Benefits

- Centralized visibility of traffic conditions across multiple intersections.
- Faster identification of congested approaches and high-density situations.
- Data-driven support for signal-timing decisions.
- Availability of historical information for traffic analysis and planning.
- Reduced dependence on manual calculation when evaluating traffic-density conditions.

Operational Feasibility Assessment:

☑ Operationally Feasible

☐ Operationally Not Feasible

Justification: The proposed interface is designed around familiar monitoring and decision-support activities. It can be introduced without major changes to the conceptual traffic-control workflow, while the prototype's software-only nature reduces operational deployment complexity.

7. Overall Feasibility Summary

Feasibility Area	Status	Justification
Technical Feasibility	Feasible	Required technologies, software and general-purpose hardware are accessible; the core system can be developed with simulated/recorded traffic data.
Operational Feasibility	Feasible	The dashboard supports existing traffic monitoring and decision-making activities and requires only basic user

		training.
Overall	Feasible	The project has a manageable scope for a four-member student team and provides a practical software-based demonstration of smart traffic management.

Overall Conclusion

Based on the technical and operational feasibility analysis, the proposed Smart Traffic Management System is feasible because it can be developed using accessible web technologies and general-purpose computing resources, while its dashboard and decision-support functions can be integrated into a realistic traffic-monitoring workflow. Limiting the initial system to simulated/recorded data and software-based signal recommendations keeps the project achievable within the available academic time and resources.

8. Expected Outcome

The completed system is expected to provide a working web-based dashboard for monitoring multiple traffic intersections, storing and visualizing traffic-density information, detecting congestion, generating adaptive signal-timing recommendations, handling simulated emergency-priority scenarios and presenting historical traffic analytics. The final prototype will demonstrate how software-based traffic analysis can support more responsive intersection management without requiring direct control of physical traffic infrastructure.

9. Future Enhancements (Optional)

- Integration with live CCTV/video streams and computer-vision-based vehicle detection.
- IoT sensor integration for real-time vehicle counts and roadside traffic measurements.
- Machine-learning models for short-term traffic prediction and proactive signal optimization.
- Integration with real traffic-signal controllers and a city-wide traffic management platform.
- Mobile notifications and advanced route recommendations for commuters and emergency services.

End of Feasibility Report